

OSTROV CHETYREHSTOLBOVOY, Russian Federation

WMO: 219650

Lat: 70.633N

Lon: 162.483E

Elev: 135

StdP: 14.62

Time Zone: 11.00 (E11)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.1	(c) -37.2	(d) -43.6	(e) 0.4	(f) -39.7	(g) -40.9	(h) 0.5	(i) -37.1	(j) 36.6	(k) -11.7	(l) 33.8	(m) -12.0	(n) 10.4	(o) 210	(p) 1.179

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	52.0	47.9	47.7	44.5	44.2	41.4	48.4	51.5	44.9	47.6	41.6	43.8	8.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
45.9	45.9	50.9	42.5	40.2	46.6	39.3	35.5	42.5	19.5	51.6	17.7	47.8	16.0	44.1	60.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.8	28.4	25.3	-41.8	58.9	2.6	6.1	-43.7	63.3	-45.2	66.9	-46.6	70.3	-48.5	74.7		
			-41.9	52.8	2.5	4.8	-43.7	56.3	-45.1	59.1	-46.5	61.8	-48.3	65.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-20.0	-17.0	-15.5	0.3	20.6	33.4	37.4	36.3	31.5	15.9	-4.4	-18.1		
	DBStd	23.91	10.29	11.65	10.78	9.76	8.12	4.50	4.47	5.08	4.41	9.25	10.92	10.63		
	HDD50	15150	2171	1875	2030	1491	911	499	393	424	555	1058	1631	2112		
	HDD65	20623	2636	2295	2495	1941	1376	949	857	888	1005	1523	2081	2577		
	CDD50	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	12.8	12.6	12.1	11.7	12.1	12.8	12.8	12.8	13.4	14.5	13.3	13.5	12.0		
	PrecAvg	3.5	0.1	0.1	0.1	0.1	0.2	0.4	0.7	0.7	0.5	0.4	0.1	0.2		
(15) Precipitation	PrecMax	5.5	0.4	0.2	0.3	0.6	0.9	1.1	1.9	2.8	1.2	0.8	0.6	0.7		
	PrecMin	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0	0.0	0.0		
	PrecStd	1.2	0.1	0.1	0.1	0.1	0.2	0.3	0.4	0.5	0.3	0.2	0.1	0.2		
	PrecStd	1.2	0.1	0.1	0.1	0.1	0.2	0.3	0.4	0.5	0.3	0.2	0.1	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	16.8	17.2	16.0	28.1	41.5	52.6	59.9	55.5	43.7	32.3	25.8	10.0			
	0.4% MCWB	15.5	16.3	15.0	26.3	38.3	47.6	53.6	50.3	40.6	30.5	25.2	9.5			
	2% DB	5.4	10.3	11.0	22.9	36.3	46.1	54.0	51.1	41.1	30.2	20.5	4.9			
	2% MCWB	4.6	9.8	9.7	21.5	33.7	41.7	48.8	48.4	39.0	28.8	19.5	4.2			
	5% DB	-1.1	4.1	4.6	19.2	33.8	43.2	48.6	47.5	39.5	28.7	17.0	1.8			
	5% MCWB	-1.7	3.5	3.9	18.3	31.9	39.3	44.9	45.4	37.8	27.0	16.2	1.2			
	10% DB	-5.2	-0.5	0.4	15.0	31.7	40.2	44.4	43.6	37.9	27.0	11.3	-2.0			
	10% MCWB	-5.5	-0.9	-0.2	14.2	30.2	37.0	41.6	41.8	36.5	25.3	10.5	-2.6			
	0.4% WB	15.7	16.5	15.0	26.9	38.3	47.5	53.9	51.1	41.4	31.1	25.1	9.4			
	0.4% MCDB	17.3	17.4	15.8	27.8	41.9	52.6	60.1	53.8	42.8	32.0	25.6	9.9			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	4.5	9.8	9.8	21.7	33.9	42.1	49.1	48.5	39.4	29.0	19.6	4.3			
	2% MCDB	5.5	10.3	11.0	22.7	35.9	45.9	53.2	51.1	40.7	30.1	20.4	5.0			
	5% WB	-1.6	3.7	4.0	18.4	32.0	39.4	45.1	45.5	38.0	27.2	16.3	1.3			
	5% MCDB	-1.0	4.2	4.6	19.4	33.5	43.0	48.8	47.7	39.3	28.4	17.1	1.8			
	10% WB	-5.6	-1.0	-0.3	14.2	30.2	37.1	41.7	42.0	36.5	25.5	10.6	-2.5			
	10% MCDB	-5.2	-0.6	0.2	15.0	31.6	39.9	44.1	43.5	37.9	26.9	11.3	-2.0			
(35) Mean Daily Temperature Range	MDBR	9.1	8.8	9.1	10.4	7.6	6.5	7.1	4.9	3.9	6.2	9.2	8.7			
	5% DB	12.8	13.9	12.5	13.0	8.5	13.1	16.5	9.8	6.5	5.2	14.4	13.4			
	5% WB	12.4	13.6	12.2	12.4	7.2	9.9	12.6	7.8	5.4	5.2	14.3	13.1			
	MCDBR	12.9	14.1	12.1	12.9	8.3	12.8	16.0	9.7	6.2	5.4	14.5	13.3			
	MCWBR	12.4	13.8	11.8	12.2	7.2	9.9	12.6	7.9	5.3	5.6	14.5	13.0			
	MCWBR	12.4	13.8	11.8	12.2	7.2	9.9	12.6	7.9	5.3	5.6	14.5	13.0			
(40) Clear-Sky Solar Irradiance	taub	N/A	0.200	0.276	0.318	0.340	0.336	0.375	0.373	0.312	0.241	N/A	N/A			
	taud	N/A	1.972	1.997	1.928	1.948	2.127	2.266	2.345	2.462	2.114	N/A	N/A			
	Ebn at Noon	0	160	220	248	260	267	251	235	221	136	0	0			
	Edh at Noon	0	12	27	41	46	40	34	28	18	10	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	102	610	1392	2010	2187	1626	967	472	133	0	0			
	RadStd	0	10	27	59	55	126	133	94	31	19	0	0			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page